Safe System of Work
Working in a Quarry

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Introduction.

The Fox Brothers Group Ltd (FG) wishes to ensure that all office and workplace environments within its operations are both managed and used in a manner that is conducive to the safety of all FG employees and other parties who may have cause to work in the quarry, for whatever reason.

There are several laws relating to the Quarry environment. In general, a Quarry, is covered in the Quarries Regulations 1999. But also covered by the Health and Safety at Work Act 1974: more specific detail is contained in the Workplace (Health, Safety and Welfare) Regulations 1992, which specify standards for the general office environment. Other legislations applicable to a quarry are the Management of Health and Safety at Work Regulations, First Aid at Work Regulations, Manual Handling Operations Regulations, Control of Substances Hazardous to Health regulations and The Control of health risks Workplace.

All H&S legislation is part of statute law, and breaches of the laws and regulations are criminal offences under the umbrella of the Health and Safety at Work etc. Act 1974. Penalties for breaches of H&S legislation can now be very severe.

Purpose.

This procedure applies to all employees of FG including contactors, agency, and sub-contracted staff. Managers at all levels are expected to take an active lead to ensure that health and safety and systems of internal control are of the highest standard and integral to the operation of the organisation. All employees and other persons are expected to have due regard for their own health and safety and that of their colleagues and other persons. If safe systems of work have been introduced, employees are expected to follow them, and any other relevant instructions.

What is a Quarry?

A quarry is a place where sand, stone or gravel is extracted, processed, and sold.

Quarries are essential to the British economy. They provide raw materials and specialist products for use in the construction industry, civil engineering projects and agriculture. Products such as cement, precast items and building materials generate export revenue.

Quarries vary greatly from small operations to large pits excavating large amounts of material and include a number of manufacturing plants. Although there are often a greater number of hazards at larger quarries, you may be at more risk at a smaller operation – depending on how well the health and safety matters are controlled at the quarry.

Every quarry must have an Operator. The Owner must appoint the Operator of the quarry. At most quarries the Owner is the Operator and is self-appointed. The Operator must be competent and have the necessary resources to operate the quarry.

Physical Hazards.

Quarry workers are often exposed to hazards by the normal, everyday events that take place in quarries. For example, excavating and processing stone or sand generally involves:

- a high number of vehicle movements,
- the operation of crushing, screening, and sizing equipment,
- work at height,
- getting in and out of large vehicles,
- risk of collapse of the sand or stone face,
- working on or near water, and
- the use of explosives.

Each of these activities involves a degree of risk.

Health Hazards.

Quarries are often noisy, dusty places and some of the work can be very physical. Such factors may expose quarry workers to an increased risk of:

- hearing loss,
- respiratory disease,
- musculo-skeletal injuries such as back injury.

Back injury can also be caused or aggravated by whole body vibration from the operation of vehicles, particularly when they run-on poor-quality surfaces.

Operations such as concrete manufacturing can bring workers into contact with chemicals and sensitisers (substances that can cause an allergic reaction over time). Such exposure increases the risk of:

- Dermatitis,
- Asthma or other respiratory problems.

Personal Protective Equipment.

You have a duty to protect your health and safety at work and that of any person affected by your actions. To comply with this duty, you must use safety equipment correctly, wear appropriate high-visibility clothing, report hazards and always adopt safe working practices and behaviours.

PPE and safety equipment:

Workers have a duty to use correctly any personal protective equipment (PPE) or safety equipment provided by their employer. Examples include hearing protection, respiratory protection, hard hats, steel toe-capped boots, seat belts in vehicles, gloves, safety harnesses and lanyards.

Workers who have a problem with the PPE or safety equipment issued must raise it with their employer to see why it is required and what alternatives may be available.

Workers handling chemicals must ensure that they are using the correct PPE as listed on the safety data sheet or on the packaging.

This equipment protects your safety and health. Always use it properly.

High-visibility clothing:

The use of high-visibility clothing greatly improves the visibility of workers in a quarry and reduces their exposure to the risk of an accident.

The high-visibility clothing provided by employers include vests, t-shirts, jackets, hats.

The operator sets the policy for the use of high-visibility clothing at the quarry, and requirements will range from waistcoats to full-body high-visibility suits.

The type of clothing worn in a quarry will also be determined by issues such as lighting in the area and conditions such as fog and snow. Quarries that operate in the hours of darkness have an increased hazard.

It is advisable for employers to provide maintenance workers and anyone who works in an area that has many moving vehicles with a full high-visibility body suit. Such suits help machinery operators to see where workers are and take the necessary precautions to protect them.

Wearing appropriate high-visibility clothing will help protect your health and safety.

Designated pedestrian routes.

Quarry operators should provide designated pedestrian routes or prohibit pedestrian access to areas where pedestrian routes cannot be provided, this is informed during the Induction packs.

When there is no pedestrian route, access should be by vehicle only, or, where access is essential, carried out within the quarry's operating procedures.

Always use the pedestrian route provided and do not enter prohibited areas.

Vehicle Checks.

Workers who operate a vehicle should carry out a daily check of that vehicle before use. This check should include testing:

- tyres,
- reversing and visibility aids,
- lights,
- access steps,

- oil and water levels,
- brakes.

Maintenance operations.

The date of the inspection and any defects should be recorded, and a plan put in place to rectify any defects. Do not use any vehicle that has a serious defect that makes it unsafe for you to operate or puts anyone else at risk.
Anyone planning to carry out maintenance work must isolate the plant and equipment before starting the work.
It may also be necessary to isolate other plant that could create a hazard if it started up. For example, if carrying out work on a crusher or screener, the feed conveyor should also be isolated.
Isolation should be physical and preferably should involve locking the electrical supply so that it cannot be re-energised unintentionally.

If carrying out maintenance on pressure systems or where there is stored energy such as hydraulic systems or work on crushers or block-making machines, the stored energy should be released first, or the equipment locked so that the stored energy cannot be released during the maintenance work.

<u>You must always isolate equipment before you start a maintenance operation.</u>

Removing and replacing guards.

Any guard or safety device removed (during maintenance work, for example) must be replaced before the equipment can be used again.

If you see a missing or faulty guard on a piece of equipment you should replace it or, if you are unable to do so, report it immediately to your supervisor.

You should never work near unguarded machinery. If you are replacing a missing guard, the machinery must first be isolated so that it cannot restart whilst the guard is being fitted.

Mobile phones and radios.

Mobile phones and radios, even hands-free devices, should not be used by anyone operating plant and machinery as they can cause a distraction that can have devastating results.

The operator and/or contractor should have a policy on the use of mobile phones and radios whilst operating plant and equipment.

<u>If you need to take or make a call or answer a radio, then the plant or machinery you are using must first be brought to a safe position.</u>

Emergency procedures.

Quarry operators and contractors should ensure that their employees are aware of the actions to take in the event of an emergency and are given the training they need.

Basic information on the emergency procedures is provided during induction training and driver/visitor induction in reception which specifies speed limits, traffic routes, firefighting, first aid and evacuation.

Traffic Management.

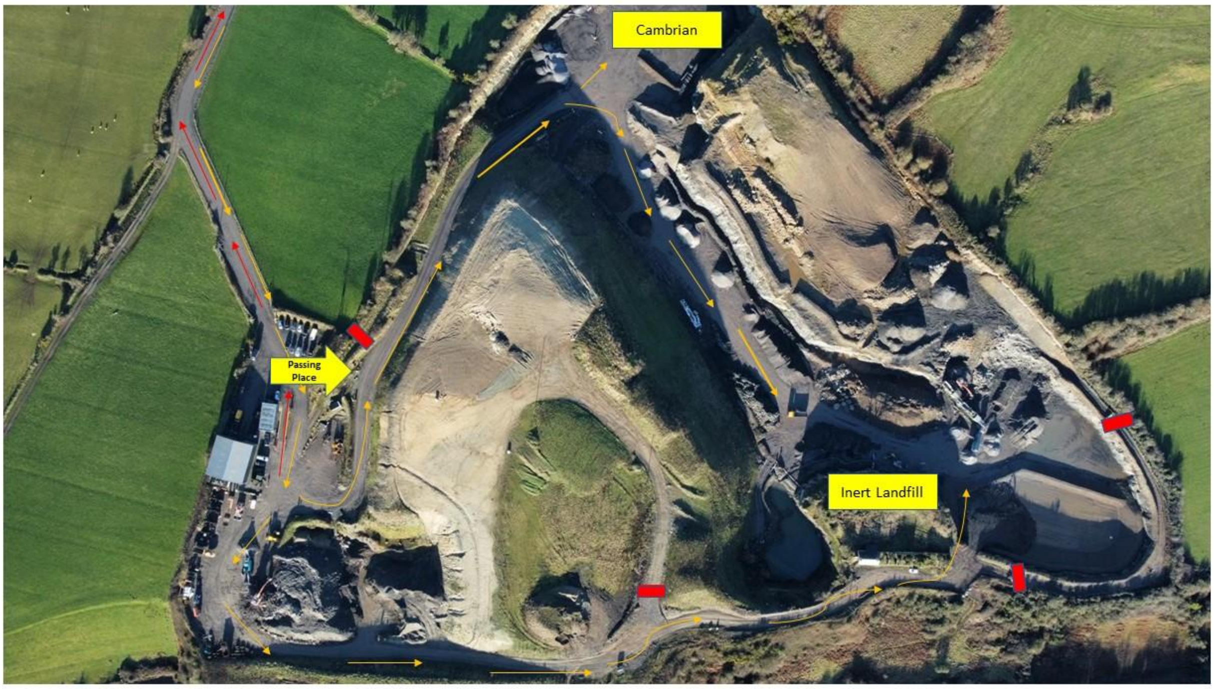
The operator has a duty to design the quarry so that vehicles can run safely.

This duty requires the operator to prepare vehicle and transport rules and to develop a traffic management plan to ensure that:

- pedestrians and vehicles are segregated wherever possible,
- roads are well maintained, without excessive gradients and tight turns,
- roads are of sufficient width to allow vehicles to pass safely, and
- edge protection is in place, where necessary.

The operator should put measures in place to eliminate reversing of vehicles, or reduce it to the lowest possible level, through the provision of adequately signed traffic routes and one-way systems.

The Traffic route can be seen in reception and will also be part of the induction given to drivers and visitors to site.



Machinery.

The operator and contractors operating their own equipment or using equipment provided by the operator for their use, shall ensure that all conveyors, crushers, sizers, and screeners, including mobile equipment, are adequately guarded at all times whilst in use.

The guarding should prevent access to all moving parts that are nip points or present a risk of entanglement for workers. This should include all drive belts and chain drives, inspection hatches, delivery and return drums, tension units and bottom belt rollers that can be accessed.

If guarding must be removed for maintenance reasons, then the equipment must be securely isolated, and the guarding replaced before the isolation is removed.

Horizontal conveyors or inclined conveyors with a walkway must have a stop pull-wire with one or more lock-out units along its length. Emergency stops and isolation points should be located at each end of the conveyor.

Mobile equipment must have emergency-stop devices located at suitable points that can be easily accessed by workers if they need to stop the machinery in an emergency.

Walkways along inclined conveyors must be maintained in a good condition and be fitted with necessary handrails and means to prevent objects falling from the walkway to the ground below.

Excavations.

Quarries occasionally require the digging of excavations for pipes and electrical services. This work can be hazardous and fatalities and serious injuries frequently occur.

Accidents can be the result of the failure of the sides of an excavation, the collapse of the excavation due to loading of materials close to it or carrying out excavations close to an existing structure or wall.

There is also potential for striking live electrical cables or gas pipelines, the excavator/ digger falling into the excavation or persons being struck by the excavator during its operation.

A safe system of work must be in place for anyone working in an excavation. Such workers must be provided with appropriate information, training and instruction and the work must be adequately supervised with the necessary control measures in place.

Design and excavation of a quarry face.

The operator has a duty to design excavations, tips and lagoons so that they can be operated safely without becoming a significant hazard.

The quarry face should be designed so that it is stable and without risk of collapse.

The maximum safe height of excavated faces is influenced by the geology and physical properties of the material being excavated; the size, height and type of machinery; and the working methods used.

A face should not be in excess of 20 metres high and local conditions or the equipment available may necessitate a much lower face.

For sand, gravel and other unconsolidated material, the maximum recommended face height is 7.5 metres, but this is again dependent upon local conditions and the equipment in use at the face.

Any excavator used at the face must:

- have a cab with falling object protection (FOP's),
- be on a suitable rock pad with an adequate rock trap,
- have adequate reach to be able to dress the face and remove any dangerous loose hanging rock.

Ponds and lagoons.

The operator must have procedures in place to prevent workers from being placed at risk of drowning at or near ponds and lagoons.

Adequate precautions must be in place to prevent persons accidentally falling into water.

Maintenance work at pumps or other equipment must be planned and, where there is a risk of a person falling into the water, flotation devices must be worn.

Rescue equipment must be in place and workers must be trained in emergency response procedures. Lone workers should not carry out tasks at or near water where there is a risk of drowning.

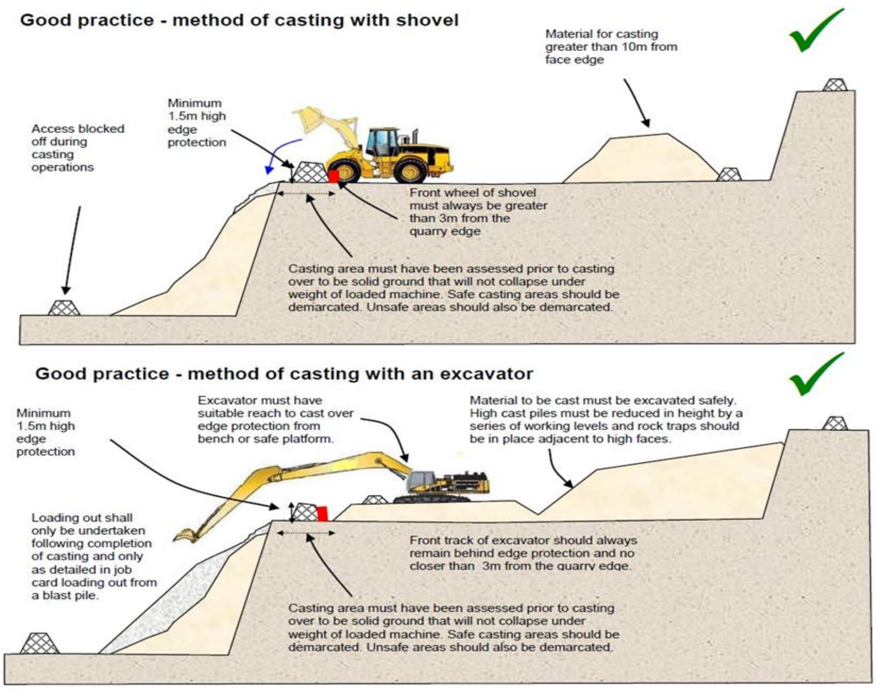
Working at the top of a Quarry face.

If workers are required to work at the top of a quarry face and there is a risk of them falling over the quarry face, then the operator must ensure that measures are in place to eliminate the risk of falling. Such measures include:

- provision of fall prevention and protection equipment – collective measures (barrier using straps and poles, etc.) or a physical barrier are preferable to individual measures (lanyards, harnesses, etc.),
- specific procedures as to how and when the fall protection is erected and taken down,
- training in the use of fall prevention equipment.

Harnesses should be used by persons erecting or dismantling the edge protection and not as fall protection during drilling and blasting.

If tipping is being carried out at the top of a quarry face, tipper trucks should not reverse to the face and tip, they should tip at a place of safety and the material should be pushed overusing a bulldozer or a loading shovel.



First Aid Arrangements.

FG are under a general duty to provide a safe place of work, with suitable arrangements for welfare. FG must ensure that there is adequate first aid provision for employees who may become ill or are injured at work. A suitable person must be appointed to take responsibility for first aid provision and maintenance of the first aid box under the Health and Safety (First

Aid) Regulations 1981 (as amended).

FG will consider the nature of activities at the workplace when determining the number and types of first aiders to appoint. As a minimum, a low-risk workplace such as a small office should have a first-aid box and a person appointed to take charge of first-aid arrangements, such as calling the emergency services if necessary.

Employees will be informed of arrangements or changes, which have been made for first aid, including the location of equipment, facilities, and appointed personnel.

Manual Handling.

Poor lifting and carrying technique contribute to manual handling related injuries of staff every year. Although there are some members of staff who lift objects on a daily basis as part of their employment, nearly all staff will lift some objects during their working week. Good technique is vital in preventing injury.

Short courses are available via Monks training that will provide correct skills, which if implemented, will help to prevent injury. If the object to be lifted is large, awkward, or heavy then an assessment should be undertaken. The first part of any assessment should consider whether the object needs to lifted at all.

Engineering methods e.g., lifting appliances, or trolleys etc., should be considered next, if this is not possible a method for manual lifting with the assistance of other staff can be used. Many people use poor techniques and have escaped injury due to their general fitness and age. However, there will be a risk of eventual injury as these conditions change. A serious back injury could cause substantial pain and be extremely debilitating.

Some tips on efficient lifting:

- is it necessary to lift the load? If not – do not!
- assess the lift and decide if help is needed.
- obtain a firm grip on the load (use gloves if necessary).
- bend at the knees not from the waist.
- use your legs not your back to thrust upwards (the leg muscles were designed for power and strength).
- keep the load near to your body.
- do not twist your spine when lifting or carrying loads.

Further advice and guidance on Manual Handling Operations can be sought through the Health and Safety Team. Also ensure you undertake your online training.

Welfare Facilities.

Welfare facilities include the provision of adequate toilet and washing facilities. FG will ensure these facilities will be in sufficient numbers, be clean, well maintained and have adequate ventilation. Hot and cold water, soap and hand drying facilities will also be in place. The provision of suitable drinking water is also a statutory requirement and will be supplied.

FG Workplace Inspections.

FG will undertake a full safety inspection of their workplace at least annually. The Quarry supervisor does weekly and monthly checks of the quarry. FG health and safety team can see these reports straight away due to the **Foxsafe** system and action any issues.

Control of Substances Hazardous to Health (COSHH).

The use of chemical substances in any environment requires strict control procedures to ensure the safety of personnel and visitors. The Control of Substances Hazardous to Health

Regulations 2002 (COSHH) requires employers to make arrangements to control the exposure of their employees to all substances which may affect their health.

FG will endeavour, wherever possible, only to use substances classified as non- hazardous. Where this is not possible and a hazardous substance cannot be substituted, strict controls will be enforced as list below:

- If using a hazardous substance, a COSHH assessment on its use must be prepared.
- Before using a new product, managers must ensure that they are in possession of the COSHH assessment. The content of which must be provided to those employees who are likely to come into contact with the product.
- Always work to the systems that are in place and follow the guidance given in the material safety data sheet (MSDS) and COSHH Assessment Form.
- Always wear the supplied PPE if necessary for the task.
- Always work in a safe and professional manner.
- DO NOT USE any hazardous substance for anything other than its intended use.
- Report to management any issues/concerns that you may have.
- Ensure hazard information is kept up to date.
- All substances that have the potential to cause harm to health must be stored safely and securely and with regards to the suppliers' recommendations.
- Ensure that employees are trained in all areas mentioned above.

FG requires all contractors (i.e., cleaners) using COSHH items within FG premises to ensure that they comply with the arrangements above.

Further advice and guidance on using COSHH can be sourced from HSQE.

Office Lone Working.

If there is a need for employees of FG to have to work on their own. There will be a risk assessment conducted by the health and safety team which will assess the risk to that person.

The main procedures that are used within lone worker are:

- To call/text Site Manager/Supervisor every hour and to report that he is leaving site report:
- Calls/texts are to be recorded by Site Manager/Supervisor.
- If one of the predetermined calls or texts is not received, Site Manager/Supervisor to send someone to site to check all is ok or alert emergency services to the site address.

Staff Inductions.

In order to secure the health and safety of all employees, FG will provide health and safety training to new employees, which will be incorporated into general induction training.

Induction training will commence on the first day of employment so that employees are familiar with basic procedures once they are at their place of work. Where this is not possible, induction training will take place as soon as possible after the employee has started work. The person responsible for this will always be the Line Manager.

The health and safety component of induction training will contain the following:

- **FG's Health and Safety Policy** — the contents of FG's policy statement will be covered in detail, including the responsibilities set out in the policy, this will enable the employee to become acquainted with the organizational arrangements.
- **Accident Reporting Procedures/First Aid** — this will cover the action to be taken when an accident has occurred, the person to be informed and where to acquire first aid treatment (this section will also cover FG's procedure as to the investigation of accidents: the reporting procedure will be explained so that the employee is aware as to what will happen when an accident occurs);
- **Fire Procedures and Precautions** — this section covers action to be taken in a fire situation and will include:
    - the location of the fire exit.
    - the assembly points.
    - the responsible person the employee must report to.
    - further instructions on the action to be taken in the event of discovering a fire.
    - what to do with machinery or processes left prior to evacuating an area.
- **Safety procedures** — items for discussion in this section could include.
    - use of display screen equipment.
    - Lone working.
    - Any additional local safety rules.

Once the induction training has been completed, a record of the training will be kept. The name of the employee, the date and subjects covered should be included.

Drivers and Visitors.

Drivers and Visitors must report to reception and be given an induction for CMP site which includes Rules on site, Site speed, Traffic Routes, and overhead power within the site. Once that is complete a site pass will be given for CMP Quarry which must be present at all times on site.

If contractors or visitors are seen acting unsafely this should be reported to HSQE so that the matter may be raised with the individual or company concerned.

Equality Assessment.

FG aims to design and implement procedural documents that meet the diverse needs of our service and workforce, ensuring that no one is placed at a disadvantage over others, in accordance with the Equality Act 2010.

Legislation and References.

- The Quarries Regulations 1999
- Health and Safety at Work etc Act 1974
- Management of Health and Safety at Work Regulations 1999
- Provision and Use of Work Equipment Regulations 1998
- Control of Substances Hazardous to Health Regulations 2002
- Control of health risks Workplace (Health, Safety and Welfare) Regulations 1992
- Manual Handling Operations Regulations 1992
- Personal Protective Equipment at Work Regulations 1992
- Control of Noise at Work Regulations 2005
- Lifting Operations and Lifting Equipment Regulations 1998
- Reporting of Injuries, Diseases and Dangerous Occurrences Regulations 2013
- Control of Vibration at Work Regulations 2005
- The Work at Height Regulations 2005
- The Workplace (Health, Safety and Welfare) Regulations 1992
- Health and Safety (First Aid) Regulations 1981
- The Health and Safety (Display Screen Equipment) Regulations 1992.
- The Electricity at Work Regulations 1989.
- Maintaining portable electric equipment in low-risk environments INDG236 (REV2).
- The Regulatory Reform (Fire Safety) Order 2005.
- The Social Security (Claims and Payments) Regulations 1979.
- The Data Protection Act 1998.